Likelihood-Based Inference with Separable Correlation Matrices

Karl Oskar Ekvall
Department of Statistics, University of Florida
k.ekvall@ufl.edu

Abstract

This paper proposes methods for likelihood-based inference in multivariate linear regressions when the correlation matrix of the responses is separable; that is, it has a Kronecker product structure, but the variances are unrestricted. The methods are enabled by a block-coordinate ascent-like algorithm with closed-form updates that strictly increases the likelihood at every iteration until convergence. In the numerical experiments, the proposed algorithm is 300–2500 times faster than a general-purpose solver, making parametric bootstrap tests of correlation and covariance separability practical. Parameters are identifiable, and standard errors can therefore be obtained from the expected Fisher information, which can be computed efficiently using the Kronecker product structure. Simulations show that the proposed estimator has lower error than both separable covariance and unrestricted estimators when the model holds, and that bootstrap tests maintain nominal size where asymptotic tests fail. An application to dissolved oxygen data from the Mississippi River demonstrates that separable correlation captures location-specific variance patterns that separable covariance cannot.

Keywords: coordinate ascent, Kronecker product, matrix normal distribution, separability, spatio-temporal data, structured covariance estimation

1 Introduction

Separable dependence structures arise in many applications, including multivariate repeated-measures data, longitudinal imaging, diffusion-tensor MRI, spatio-temporal environmental

data, mortality tables, genomics, and international trade data (Naik and Rao, 2001; George and Aban, 2015; Basser and Pajevic, 2003; Zapata et al., 2022; Dutilleul and Pinel-Alloul, 1996; Fosdick and Hoff, 2014; Allen and Tibshirani, 2012; Volfovsky and Hoff, 2015). Often, the data are matrix-valued, and then separability says dependence patterns among rows are similar across columns, and vice versa.

To be more precise, suppose there are n independent $r \times c$ response matrices $Y_1, \ldots, Y_n$; an extension to tensor-valued data is discussed later. Each Y_i has elements Y_{ijk}, $j \in \{1, \ldots, r\}$ and $k \in \{1, \ldots, c\}$. The focus is on inference on the covariance structure for observations in the same matrix Y_i, which is characterized by the covariance matrix of the vector of all responses in Y_i. Let $\mathrm{vec}(Y_i) = [Y_{i11}, Y_{i12}, \ldots, Y_{irc}]^\top \in \mathbb{R}^{rc}$ denote this vector, and $\Sigma = \mathrm{cov}\{\mathrm{vec}(Y_i)\}$. The matrix Σ is separable if $\Sigma = \Sigma_2 \otimes \Sigma_1$ for some positive definite $\Sigma_1 \in \mathbb{R}^{r \times r}$ and $\Sigma_2 \in \mathbb{R}^{c \times c}$. Equivalently, Σ is separable if every column of Y_i has covariance matrix proportional to Σ_1, the constant of proportionality being the corresponding diagonal element of Σ_2, and every row of Y_i has covariance matrix proportional to Σ_2.

In some applications, covariance separability is too restrictive, yet estimating an unstructured covariance matrix is also inappropriate, either because it ignores the structure of the data or because the number of parameters is large relative to the sample size. It may then be useful to assume that the correlation matrix is separable but that the variances need not be. Formally,

$$\Sigma = D(C_2 \otimes C_1)D, \tag{1}$$

where $D = \mathrm{diag}(d_1, \ldots, d_{rc}) = \mathrm{diag}(\sqrt{\Sigma_{11}}, \ldots, \sqrt{\Sigma_{rc}})$ is a diagonal matrix of standard deviations, and C_1 and C_2 are correlation matrices of dimensions $r \times r$ and $c \times c$, respectively. The difference from separable covariance is that D need not be separable. Consequently, the number of parameters is larger than with separable covariance, $rc + r(r-1)/2 + c(c-1)/2$ compared to $r(r+1)/2 + c(c+1)/2 - 1$, the last -1 coming from an identifiability constraint such as $[\Sigma_2]_{11} = 1$. Thus, as r and c grow, both parameter counts are $O(r^2 + c^2)$ and are much smaller than the $O(r^2c^2)$ parameters needed for an unstructured covariance matrix.

While separability of correlations has been considered in the spatio-temporal statistics literature (see Gneiting et al., 2006, for example), the likelihood-based methods studied here have not. A main contribution is a block coordinate ascent algorithm with rescaling for maximizing the likelihood. Thanks to closed-form updates which are fast to compute, the algorithm is very efficient, often several orders of magnitude faster than off-the-shelf solvers. It is also stable, increasing the likelihood at every iteration until it reaches a stationary point. A full theoretical analysis of the optimization problem, similar to those available for the

separable covariance model (Wiesel, 2012; Roś et al., 2016; Soloveychik and Trushin, 2016; Drton et al., 2021; Derksen and Makam, 2021), is well beyond the scope of the paper, but extensive numerical evidence suggests the algorithm is guaranteed to converge under mild conditions (Section 4).

Unlike the separable covariance model, where the component matrices are only identified up to a scalar factor, the parameters of the separable correlation model are identifiable. Standard errors for individual entries of C_1, C_2, and D can therefore be obtained from the expected Fisher information. The computational efficiency of the algorithm also makes parametric bootstrap likelihood ratio tests of separability practical; simulations show that such tests maintain nominal size in settings where asymptotic tests fail.

2 Likelihood-based inference

2.1 Maximum likelihood algorithm

In some applications, including the data example in Section 3, every element of Y_i is affected by the same vector of predictors $x_i \in \mathbb{R}^p$, $i \in \{1, \ldots, n\}$. This paper treats this setting for simplicity, but the methods extend to cases where there are different predictors for different elements.

Assume for now, with Σ as in (1) and $\mathcal{B} \in \mathbb{R}^{p \times rc}$ a matrix of regression coefficients, that

$$\operatorname{vec}(Y_i) \sim \mathcal{N}_{rc}(\mathcal{B}^\top x_i, \Sigma), \quad i \in \{1, \ldots, n\}. \tag{2}$$

When $n > p$, because each element has the same vector of predictors, the maximum likelihood estimate of $\mathcal{B}$ is the least squares estimate $\hat{\mathcal{B}} = (\sum_{i=1}^n x_i x_i^\top)^{-1} \sum_{i=1}^n x_i \operatorname{vec}(Y_i)^\top$. Let $E_i \in \mathbb{R}^{r \times c}$ be the matrix of residuals defined by $\operatorname{vec}(E_i) = \operatorname{vec}(Y_i) - \hat{\mathcal{B}}^\top x_i$. Then the profile log-likelihood $\ell_n(D, C_1, C_2)$ is, up to an additive constant,

$$-\frac{n}{2} \log |D(C_2 \otimes C_1)D| - \frac{1}{2} \sum_{i=1}^n \operatorname{vec}(E_i)^\mathsf{T} \{D(C_2 \otimes C_1)D\}^{-1} \operatorname{vec}(E_i). \tag{3}$$

To ensure the function is well-defined in the expressions that follow, let us identify D with the vector $(d_1, \ldots, d_{rc})$ of its diagonal elements and extend ℓ_n to $\mathbb{R}^{rc} \times \mathbb{R}^{r \times r} \times \mathbb{R}^{c \times c}$ by setting $\ell_n(d_1, \ldots, d_{rc}, C_1, C_2) = -\infty$ when at least one of D, C_1, and C_2 are not positive definite.

Then, maximum likelihood corresponds to the constrained optimization problem

$$\max_{(d_1,\dots,d_{rc})\in\mathbb{R}^{rc}, C_1\in\mathbb{R}^{r\times r}, C_2\in\mathbb{R}^{c\times c}} \ell_n(d_1,\dots,d_{rc},C_1,C_2) \tag{4}$$
$$\text{subject to } d_1,\dots,d_{rc}>0, C_1 \text{ and } C_2 \text{ are correlation matrices.}$$

The proposed approach is a block-coordinate ascent with rescaling, iteratively updating D, C_1, and C_2 until convergence. Letting (k) denote iterations, the strategy is to first find quantities $\check{D}^{(k+1)}$, $\check{C}_1^{(k+1)}$, and $\check{C}_2^{(k+1)}$ which increase the log-likelihood but which need not be in the feasible set; that is, they need not satisfy the constraints in (4). Then those quantities are rescaled to get iterates $D^{(k+1)}$, $C_1^{(k+1)}$, and $C_2^{(k+1)}$ that are in the feasible set and give the same value of the log-likelihood.

At the kth iteration, compute, for $j\in\{1,\dots,rc\}$,

$$\begin{aligned}
\check{d}_j^{(k+1)} &= \operatorname{argmax}_{d_j>0} \ell_n(\check{d}_1^{(k+1)},\dots,\check{d}_{j-1}^{(k+1)},d_j,d_{j+1}^{(k)},\dots d_{rc}^{(k)},C_1^{(k)},C_2^{(k)});\\
\check{C}_1^{(k+1)} &= \operatorname{argmax}_{C_1\in\mathbb{R}^{r\times r}} \ell_n(\check{D}^{(k+1)},C_1,C_2^{(k)});\\
\check{C}_2^{(k+1)} &= \operatorname{argmax}_{C_2\in\mathbb{R}^{c\times c}} \ell_n(\check{D}^{(k+1)},\check{C}_1^{(k+1)},C_2),
\end{aligned}$$

where $\check{D}^{(k+1)}=\operatorname{diag}(\check{d}_1^{(k+1)},\dots,\check{d}_{rc}^{(k+1)})$. These updates have closed form expressions, which leads to a computationally efficient algorithm. For brevity, the solutions are stated here; details are in the Appendix. Define, for $j\in\{1,\dots,rc\}$ and $k\in\{0,1,2,\dots\}$,

$$a_j^{(k+1)} = \left\{\left(C_2^{(k)}\otimes C_1^{(k)}\right)^{-1}\tilde{D}_{(-j)}^{(k+1)}S_n\right\}_{jj},$$

where $S_n=\sum_{i=1}^n \operatorname{vec}(E_i)\operatorname{vec}(E_i)^\mathsf{T}$ and $\tilde{D}_{(-j)}^{(k+1)}=\operatorname{diag}(1/\check{d}_1^{(k+1)},\dots,1/\check{d}_{j-1}^{(k+1)},0,1/d_{j+1}^{(k)},\dots,1/d_{rc}^{(k)})$. Also letting $b_j^{(k)}=4(C_2^{(k)}\otimes C_1^{(k)})_{j,j}^{-1}\{S_n\}_{j,j}$, the maximizer is

$$\check{d}_j^{(k+1)} = \frac{a_j^{(k+1)}+\sqrt{(a_j^{(k+1)})^2+nb_j^{(k)}}}{2n}. \tag{5}$$

Maximizing in C_1 and C_2 is more straightforward. Define $\check{E}_i^{(k+1)}$ by $\operatorname{vec}(\check{E}_i^{(k+1)}) = (\check{D}^{(k+1)})^{-1}\operatorname{vec}(E_i)$. Then

$$\check{C}_1^{(k+1)} = \frac{1}{nc}\sum_{i=1}^n \check{E}_i^{(k+1)}(C_2^{(k)})^{-1}(\check{E}_i^{(k+1)})^\mathsf{T}. \tag{6}$$

and

$$\check{C}_2^{(k+1)} = \frac{1}{nr}\sum_{i=1}^{n}(\check{E}_i^{(k+1)})^{\mathsf{T}}(\check{C}_1^{(k+1)})^{-1}\check{E}_i^{(k+1)}. \tag{7}$$

The last step of the iteration is the rescaling. Define the matrix of standard deviations $\check{D}_1^{(k+1)} = \{I_r \circ \check{C}_1^{(k+1)}\}^{1/2}$, where $\circ$ is the elementwise product; and, similarly, $\check{D}_2^{(k+1)} = \{I_c \circ \check{C}_2^{(k+1)}\}^{1/2}$. Then the updates are

$$D^{(k+1)} = (\check{D}_2^{(k+1)} \otimes \check{D}_1^{(k+1)})\check{D}^{(k+1)}; \tag{8}$$

$$C_1^{(k+1)} = (\check{D}_1^{(k+1)})^{-1}\check{C}_1^{(k+1)}(\check{D}_1^{(k+1)})^{-1}; \tag{9}$$

$$C_2^{(k+1)} = (\check{D}_2^{(k+1)})^{-1}\check{C}_2^{(k+1)}(\check{D}_2^{(k+1)})^{-1}. \tag{10}$$

Routine calculations show these updates are indeed in the feasible set. In particular, $C_1^{(k+1)}$ and $C_2^{(k+1)}$ are correlation matrices by construction, and $D^{(k+1)}$ is a diagonal matrix of positive entries. Moreover, the rescaling does not change the value of the log-likelihood since the resulting covariance matrix is the same:

$$D^{(k+1)}(C_2^{(k+1)} \otimes C_1^{(k+1)})D^{(k+1)} = \check{D}^{(k+1)}(\check{C}_2^{(k+1)} \otimes \check{C}_1^{(k+1)})\check{D}^{(k+1)}.$$

The full algorithm is given in Algorithm 1. The following theorem provides theoretical support for the algorithm. The theorem does not assume normality; that is, the convergence properties of the algorithm hold even if fitting the model to non-normal data.

Theorem 2.1. *If $n-p \geq \max(r,c)$ and the elements of each Y_i have a joint continuous distribution, then the following hold for Algorithm 1 almost surely: (i) $\check{d}_1^{(k+1)}, \ldots, \check{d}_{rc}^{(k+1)}, \check{C}_1^{(k+1)}, \check{C}_2^{(k+1)}$ are unique solutions to their respective optimization problem for every k; (ii) $d_1^{(k)}, \ldots, d_{rc}^{(k)}, C_1^{(k)}, C_2^{(k)}$ are in the feasible set for every k; and (iii) every iteration strictly increases the log-likelihood, unless the current iterate is a stationary point, in which case the algorithm terminates.*

In practice, the sample size condition is not binding: if the algorithm completes without encountering an indefinite update, then every iterate is in the feasible set and the log-likelihood strictly increased at every iteration, regardless of whether $n - p \geq \max(r, c)$ holds.

For small sample sizes, (4) need not have a solution. The following proposition provides a sample size condition which ensures a solution. Numerical evidence in Section 4 suggests quite a bit smaller sample sizes are enough for a unique solution to exist, and for Algorithm 1 to converge to it.

Proposition 2.2. *When $n > p + rc$, the optimization problem* (4) *has a solution with probability one.*

Algorithm 1 Maximum likelihood with separable correlation

Input: $C_1^{(0)} = I_r, C_2^{(0)} = I_c, D^{(0)} = I_{rc}, \ell_n^{(0)} = -\infty, k = 0$, `tol` > 0
while $|\ell_n^{(k)} - \ell_n^{(k-1)}|/\ell_n^{(k-1)} > |$`tol`$|$ **do**
  for $j = 1, \ldots, rc$ **do**
    Compute $\check{d}_j^{(k+1)}$ as in (5)
  end for
  Compute $\check{C}_1^{(k+1)}$ as in (6)
  Compute $\check{C}_2^{(k+1)}$ as in (7)
  Rescale to get $D^{(k+1)}$, $C_1^{(k+1)}$, and $C_2^{(k+1)}$ as in (8)–(10)
  Compute $\ell_n^{(k+1)} = \ell_n(D^{(k+1)}, C_1^{(k+1)}, C_2^{(k+1)})$
  Update $k \leftarrow k + 1$
end while

2.2 Standard errors

The parameters of the separable correlation model are identifiable (Proposition 2.3), unlike the separable covariance model where Σ_1 and Σ_2 are only identified up to a scalar factor. Standard errors for individual entries of C_1, C_2, and D can therefore be obtained from the expected Fisher information, which can be computed efficiently using the Kronecker product structure (see the Appendix). The observed information could also be used, but modern theory for inference on variances and covariances in linear models suggests that the expected information is more reliable, especially when the true parameters are close to the boundary of the parameter space (Ekvall and Bottai, 2022, 2026; Zhang et al., 2025).

Proposition 2.3. *For any two sets of parameters $(D, C_1, C_2) \neq (D', C_1', C_2')$, it holds that $D(C_2 \otimes C_1)D \neq D'(C_2' \otimes C_1')D'$ and, consequently, the parameters are identifiable in any family of distributions determined by the covariance matrix Σ.*

2.3 Testing separable correlation

Natural hypotheses to test are (a) H_0: Separable covariance v. H_A: Separable correlation, and (b) H_0: Separable correlation v. H_A: Unrestricted covariance. For both (a) and (b), parametric bootstrap likelihood ratio tests are recommended since, as the simulation results

show, classical likelihood ratio tests are in general anti-conservative. Test (a) requires $n \geq p + rc$ since otherwise the log-likelihood under the alternative is unbounded, implying the test would reject with probability one.

The parametric bootstrap proceeds as follows. Using the original dataset, compute $\hat{\Sigma}_0$ and $\hat{\mathcal{B}}$, the maximum likelihood estimates under the null hypothesis; for test (a) $\hat{\Sigma}_0 = \hat{\Sigma}_{cov}$ and for test (b) $\hat{\Sigma}_0 = \hat{\Sigma}_{cor}$. Generate B independent datasets, each with n observations, from the model under H_0 with $\Sigma = \hat{\Sigma}_0$ and $\mathcal{B} = \hat{\mathcal{B}}$. For each simulated dataset $j \in \{1, \ldots, B\}$, compute the maximum likelihood estimates $\hat{\Sigma}^j$ and $\hat{\mathcal{B}}^j$ under the alternative, and the maximum likelihood estimate $\hat{\Sigma}_0^j$ under the null. For a level α test, reject the null hypothesis if the likelihood ratio $L(\hat{\Sigma}, \hat{\mathcal{B}})/L(\hat{\Sigma}_0, \hat{\mathcal{B}})$ obtained from the original data exceeds the $(1-\alpha)$th empirical quantile of $\{\xi_1, \ldots, \xi_B\}$, where $\xi_j = L(\hat{\Sigma}^j, \hat{\mathcal{B}}^j)/L(\hat{\Sigma}_0^j, \hat{\mathcal{B}}^j)$ and $L(\mathcal{B}, \Sigma)$ denotes the multivariate normal likelihood for $\mathrm{vec}(Y_1), \ldots, \mathrm{vec}(Y_n)$ with respective means $\mathcal{B}^\mathsf{T} x_i$ $(i = 1, \ldots, n)$ and common covariance matrix Σ.

2.4 Regularized estimation

When n is small relative to r, c, or p, the iterates in Algorithm 1 may not exist or, even if they do, the maximum likelihood estimator may not exist or have high variance. A natural remedy is a regularized estimator that maximizes

$$\ell_n(D, C_1, C_2) - \frac{\lambda}{2}\,\mathrm{tr}(\Sigma^{-1}), \quad \lambda \geq 0. \tag{11}$$

Because Σ is positive definite, $\mathrm{tr}(\Sigma^{-1})$ is the nuclear norm of Σ^{-1}; the trace is the sum of eigenvalues, which are the singular values. This penalty is convenient because it is invariant under the rescaling step in Algorithm 1; monotone increase of the penalized objective then follows by the same argument as Theorem 2.1(iii). Second, the updates for D, C_1, and C_2 retain closed-form solutions (Appendix). The penalty parameter λ can be chosen by cross-validation or an information criterion.

3 Data example

The method is illustrated using dissolved oxygen concentration data from the Mississippi River, collected by the US Army Corps of Engineers' Upper Mississippi River Restoration Program Long Term Resource Monitoring element (Johnson and Hagerty, 2008). The mean structure uses cubic spline regressors in the year index to remove systematic trends; since

the choice of predictors does not affect estimation of Σ, it is not discussed further. The data consist of $n = 21$ years (1994 – 2002, 2004 – 2015) of quarterly measurements from $r = 16$ areas of the Upper Mississippi River. Observations from winter are excluded since water is typically mostly frozen in the northernmost sampling areas in winter, so $c = 3$.

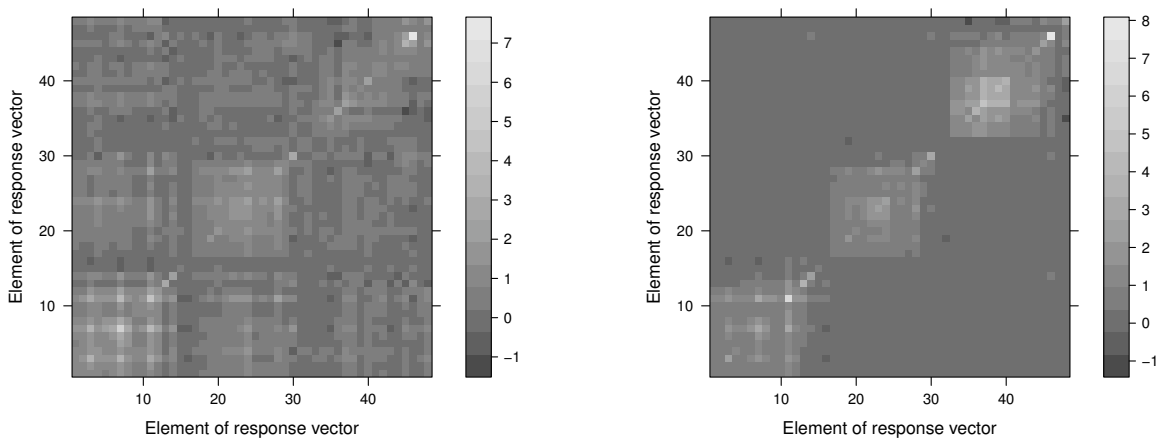
Dissolved oxygen measurements are treated as independent among years. In the raw data there are several measurements for every year, season, and area. For convenience, the sample means of these measurements are modeled. Let $Y_{i,j,k}$ ($i = 1, \ldots, n$, $j = 1, \ldots, 16$, $k = 1, 2, 3$) denote the sample mean of the dissolved oxygen measurement in season k, area j, and year i. For the sample means data, the separable correlation model says that the correlations between mean dissolved oxygen concentrations in spring, summer, and fall are the same for all areas. Additionally, the correlations between mean concentrations from different areas are the same in all seasons. Without imposing additional restrictions, other than independence among years, the covariance matrix Σ would still be of size 48×48 in this example, and there are only $n = 21$ years of data. Thus, in this example, correlation separability is motivated both as a parsimonious parameterization enabling maximum likelihood estimation, and as a reflection of the spatio-temporal structure of the data. It is also of scientific interest to examine whether, in fact, the separable covariance model can be applied to these data.

The parametric bootstrap test of H_0: Separable covariance v. H_A: Separable correlation with $B = 10000$ yields a p-value less than 0.0001. Each bootstrap sample requires fitting both models, so the bootstrap involves 20000 fits in total. Using Algorithm 1, this takes approximately two minutes. With a general-purpose optimizer (BFGS), a single fit takes roughly 2.8 seconds, so the same bootstrap would take over 15 hours. Correlation separability cannot be tested against an unstructured covariance matrix here since $n = 21 < 48 = rc$.

The estimate of the season correlation matrix C_2 has off-diagonal entries -0.02, 0.10, and 0.08, with standard errors 0.055, 0.054, and 0.054 from the expected information (Section 2.2). A Wald test of $H_0\colon C_2 = I_3$ gives a test-statistic value of 5.96 ($df = 3$, $p = 0.11$), indicating only weak evidence against the null hypothesis of no seasonal correlation.

The estimated variance of response element 46 is notably large (Fig. 2). This element corresponds to a sampling location with deep water and low flow, measured in fall when flow is generally lowest (B. Gray, personal communication, Jan 2018). These conditions may cause larger year-to-year swings in dissolved oxygen, offering a physical explanation. This finding is inconsistent with separable covariance, which requires any season effect on variance to be uniform across locations. The variance estimates in Fig. 2 reveal other elements where the difference between models is even greater; for example, separable correlation indicates

particularly large variation at sampling location 11 in spring. Such location-specific findings would not be detected under separable covariance.



(a) Residual covariance matrix (b) Estimate with separable correlation

Figure 1: Estimated covariances for dissolved oxygen data. Opacity indicates covariance.

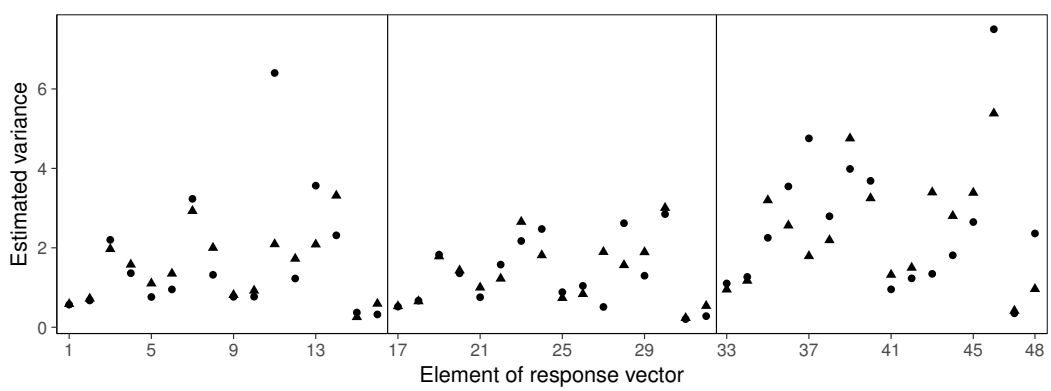


Figure 2: Estimated variances for dissolved oxygen data. Dots and triangles correspond to estimates from separable correlation and separable covariance models, respectively. The three panels are, from left to right, spring, summer, and fall.

4 Simulations

The parameters for the simulations are taken from the data example, with $r = 16$ areas, $c = 3$ seasons, and $x_i = 1$, $\beta = 0$. Under the separable correlation data generating process, C_1, C_2,

and D are the maximum likelihood estimates from the separable correlation model fit to the data. Under the separable covariance data generating process, Σ_1 and Σ_2 are the maximum likelihood estimates from the separable covariance model. The algorithm uses the default convergence tolerance of the `sepcor` package (10^{-8}) and a maximum of 1000 iterations.

Figure 3 shows average spectral norm errors $m^{-1}\sum_{j=1}^{m}\|\hat{\Sigma}^j - \Sigma\|$ across $m = 200$ simulation runs for the separable correlation, separable covariance, and unrestricted estimators. Under the separable correlation data generating process, the separable correlation estimator has lower error than the alternatives at all sample sizes. The separable covariance estimator has higher error that decreases slowly, due to persistent bias from incorrectly constraining D. Under the separable covariance data generating process, separable covariance naturally performs better, but the difference to separable correlation is relatively small. When covariance is not separable, the unrestricted estimator, available only when $n > q = 48$, performs better than the separable covariance estimator at large sample size.

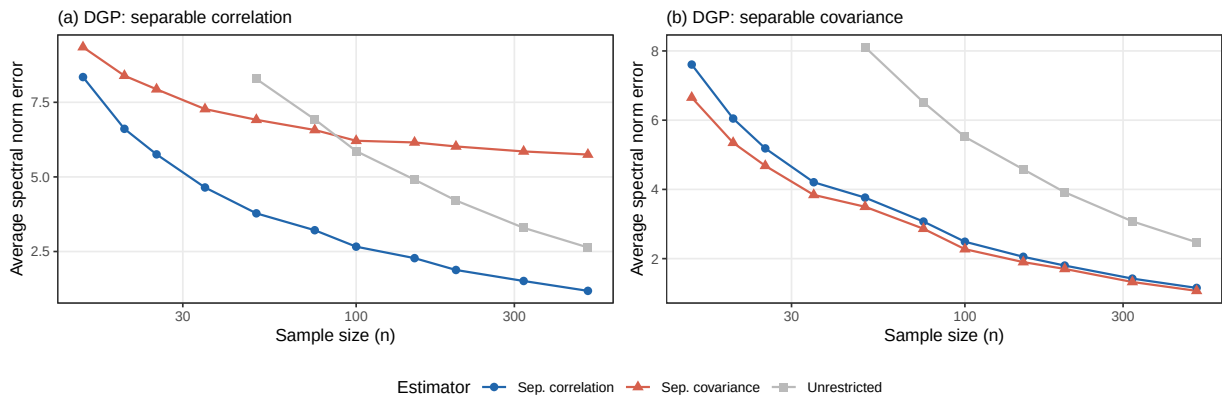


Figure 3: Average spectral norm error vs. sample size for three estimators, under two data generating processes. Data are generated with parameters set to the estimates from the separable covariance (Panel (a)) or correlation model (Panel (b)) in the data example.

Figure 4 shows rejection rates under the fitted DGPs from the data example ($r = 16$, $c = 3$). The bootstrap maintains close to nominal size in both panels. In panel (a), the asymptotic test over-rejects at small n, exceeding 0.40 at $n = 15$. In panel (b), the asymptotic test rejects in every replication: with n close to $q = 48$, the sample covariance overfits the data and inflates the test statistic far beyond the χ^2_{1005} quantiles. The bootstrap draws from the same small-sample distribution and controls size.

For the separable covariance model, there has been substantial research on determining which dimensions (n, r, c) ensure, almost surely, (i) a bounded likelihood, (ii) existence of a

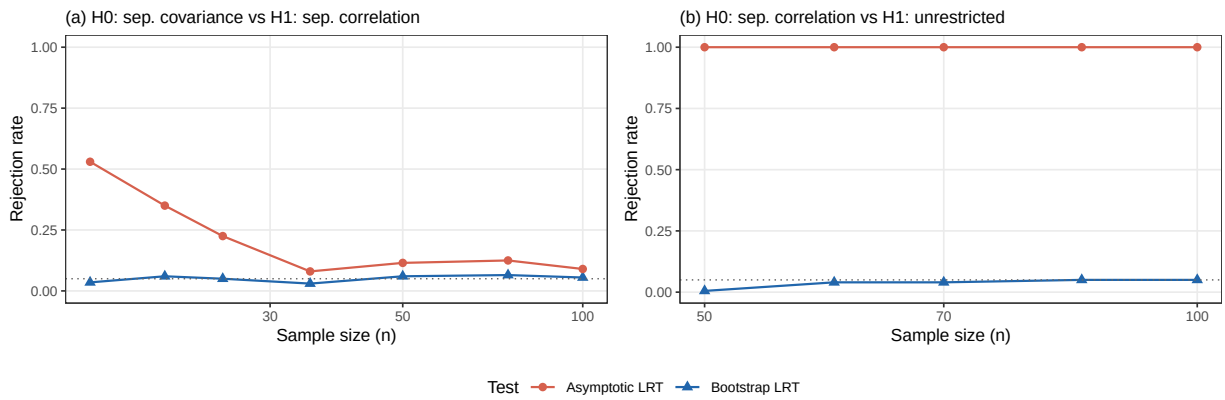


Figure 4: Sizes for the bootstrap and asymptotic likelihood ratio tests ($m = 200$ replications, $\alpha = 0.05$). Data are generated with parameters set to the estimates from the separable covariance (Panel (a)) or correlation model (Panel (b)) in the data example. The dotted line marks $\alpha = 0.05$.

maximum likelihood estimator, and (iii) uniqueness of the maximum likelihood estimator. Here, these questions are explored empirically for the separable correlation model. Data are generated from a separable covariance model with autoregressive correlation matrices $[C_1]_{jk} = [C_2]_{jk} = 0.5^{|j-k|}$ and $D = I_{rc}$. For each (r, c), 50 datasets are simulated and the algorithm is run from 20 random starting values. Let $\hat{n}_1$ be the smallest n for which all 20 starts converge for all 50 datasets, possibly to different parameter vectors. Let also $\hat{n}_2$ be the smallest n for which the 20 converged solutions for each dataset agree within 10^{-4} in maximum absolute parameter difference. Then $\hat{n}_1 \leq \hat{n}_2$, the former being an empirical analogue of almost sure existence and the latter of almost sure uniqueness.

Table 1 reports these thresholds alongside the exact theoretical bounds for separable covariance from Derksen and Makam (2021). For the generic case $r \neq c$ with $\gcd(r, c) = 1$, these simplify to $n_1 = n_2 = \lceil r/c + c/r \rceil$; for $r = c$, $n_1 = 1$ and $n_2 = 3$. The separable correlation thresholds are consistently larger, reflecting the additional rc variance parameters. For every considered $n \geq \hat{n}_2$, all 20 starting values converge to the same parameter vector, providing empirical evidence that the maximum likelihood estimator is unique whenever it exists. When $\lambda > 0$, the penalized estimator converges even when the unpenalized algorithm fails; for example, with $r = 2$, $c = 9$, and $n = 5$, the penalized algorithm with $\lambda = 1$ always converges.

To assess the coverage of Wald confidence intervals based on the expected Fisher information, 95% intervals were computed for each off-diagonal entry of C_1 and C_2 across 500

simulation runs with $r = c = 5$, $D = I_{25}$, $[C_1]_{jk} = [C_2]_{jk} = 0.6^{|j-k|}$, and $n \in \{50, 100, 320\}$. Average empirical coverage rates range from 0.94 to 0.96 across all settings, within two Monte Carlo standard errors of the nominal level even at $n = 50$.

Table 1: Sample size thresholds for existence and uniqueness. For separable covariance, n_1 and n_2 are the exact theoretical bounds from Derksen and Makam (2021). For separable correlation, $\hat{n}_1$ and $\hat{n}_2$ are empirical thresholds based on 50 simulated datasets with 20 random starts each.

		Sep. covariance		Sep. correlation	
r	c	n_1	n_2	$\hat{n}_1$	$\hat{n}_2$
2	2	1	3	4	5
2	3	2	2	3	5
2	6	3	4	6	6
2	9	5	5	8	8
3	3	1	3	4	4
3	6	2	3	5	6
5	5	1	3	5	5
5	6	2	2	5	5
5	10	2	3	7	7

Table 2 compares computing times for the proposed algorithm against BFGS, where both use the same Kronecker-structured log-likelihood implemented in `RcppArmadillo`. Data are generated from a separable correlation model with $[C_1]_{jk} = 0.5^{|j-k|}$, $[C_2]_{jk} = 0.4^{|j-k|}$, and $D = \mathrm{diag}(d_1, \ldots, d_{rc})$ with d_j equally spaced from 0.5 to 2. The proposed algorithm is 333–2504 times faster across the settings considered, with the advantage growing in $q = rc$.

Table 2: Mean computing time over 5 replications.

r	c	n	sepcor (ms)	BFGS (s)	Speedup
5	6	30	0.58	0.19	333×
8	8	50	1.55	0.77	499×
6	10	40	1.41	0.78	555×
10	10	50	2.90	2.10	723×
8	16	50	4.11	4.90	1191×
12	12	50	4.99	3.01	604×
16	16	50	14.23	35.63	2504×

All simulations and timing comparisons were run sequentially (single-threaded) on an Apple M1 Max (3.2 GHz) with 32 GB of RAM, using R 4.5.2.

5 Final remarks

The proposed algorithm extends naturally to tensor-valued responses, where the correlation matrix factors as a Kronecker product of $K > 2$ matrices, $C_K \otimes \cdots \otimes C_1$. The coordinate ascent structure carries over: cycle through each C_k, update over the larger space of positive definite matrices, and rescale to a correlation matrix. The nuclear norm penalty (11) generalizes similarly.

If the elements of each Y_i need not have the same predictors, say $\mathbb{E}(Y_{ijk}) = x_{ijk}^{\mathsf{T}}\beta$ for $x_{ijk}, \beta \in \mathbb{R}^p$, then the maximum likelihood estimator for β in general depends on the maximum likelihood estimator for Σ. The proposed algorithm can then be adapted by including an additional step to update the regression coefficients β after updating D, C_1, and C_2. The update for β has a familiar, generalized least squares-type closed-form solution, and the monotone increase property of the algorithm is preserved by the same argument as in Theorem 2.1(iii).

It is straightforward to impose restrictions such as $C_1 = I_r$ by skipping the update of C_1, effectively fixing $C_1^{(k+1)} = I_r$ at every iteration. This would be a useful model for spatial data with no temporal correlation, for example. Such a restriction can be tested in the same way as described in Section 2.3.

In the data example, further years of sampling would permit formal testing of the validity of the assumed separability against an unstructured covariance matrix. Restrictions on either the spatial or temporal autocorrelation could also be incorporated in such a study, building on the work of Szczepańska-Álvarez et al. (2017).

The computational advantages of the proposed algorithm are in line with the best algorithms for the separable covariance model: Lu and Zimmerman (2005) report that the flip-flop algorithm for the separable covariance model, which is also a block-coordinate descent algorithm (Dutilleul, 1999) is 50 to 5000 times faster than Newton–Raphson for separable covariance, and the gap grows with dimension.

Theoretical results on the existence and uniqueness of the maximum likelihood estimator for the separable correlation model are an avenue for future research. For example, it would be of interest to explore whether the likelihood is pathwise concave along some appropriate path, as it is along geodesics in the space of positive definite matrices for the separable covariance model.

Further work on regularized estimation may also be of interest. In addition to exploring other penalties than the one considered here, it may be possible to extend the core shrinkage of Hoff et al. (2023) to the separable correlation model.

Acknowledgments

The author thanks Brian Gray, Aaron Molstad, and Daniel Eck for discussions and suggestions that led to the improvement of this article. Ekvall gratefully acknowledges support by the American–Scandinavian Foundation.

Supplementary material

The supplementary material includes proofs of propositions, additional simulation results, more details regarding the data example, and convergence diagnostics for the proposed algorithm.

References

Allen, G. I. and Tibshirani, R. (2012). Inference with transposable data: Modelling the effects of row and column correlations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 74(4):721–743.

Basser, P. J. and Pajevic, S. (2003). A normal distribution for tensor-valued random variables: Applications to diffusion tensor MRI. *IEEE Transactions on Medical Imaging*, 22(7):785–794.

Derksen, H. and Makam, V. (2021). Maximum Likelihood Estimation for Matrix Normal Models via Quiver Representations. *SIAM Journal on Applied Algebra and Geometry*, 5(2):338–365.

Drton, M., Kuriki, S., and Hoff, P. (2021). Existence and uniqueness of the Kronecker covariance MLE. *The Annals of Statistics*, 49(5):2721–2754.

Dutilleul, P. and Pinel-Alloul, B. (1996). A doubly multivariate model for statistical analysis of spatio-temporal environmental data. *Environmetrics (London, Ont.)*, 7(6):551–565.

Dutilleul, Pierre. (1999). The MLE algorithm for the matrix normal distribution. *Journal of Statistical Computation and Simulation*, 64(2):105–123.

Ekvall, K. O. and Bottai, M. (2022). Confidence regions near singular information and boundary points with applications to mixed models. *The Annals of Statistics*, 50(3):1806–1832.

Ekvall, K. O. and Bottai, M. (2026). Uniform inference in linear mixed models. *Biometrika*, 113(1):asaf079.

Fosdick, B. K. and Hoff, P. D. (2014). Separable factor analysis with applications to mortality data. *The Annals of Applied Statistics*, 8(1):120–147.

George, B. and Aban, I. (2015). Selecting a separable parametric spatiotemporal covariance structure for longitudinal imaging data. *Statistics in Medicine*, 34(1):145–161.

Gneiting, T., Genton, M. G., and Guttorp, P. (2006). Geostatistical space-time models, stationarity, separability, and full symmetry. *Monographs On Statistics and Applied Probability*, 107:151.

Hoff, P., McCormack, A., and Zhang, A. R. (2023). Core shrinkage covariance estimation for matrix-variate data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 85(5):1659–1679.

Johnson, B. L. and Hagerty, K. H. (2008). Status and trends of selected resources in the Upper Mississippi River system. Techreport, U.S. Geological Survey, Upper Midwest Environmental Sciences Center.

Lu, N. and Zimmerman, D. L. (2005). The likelihood ratio test for a separable covariance matrix. *Statistics & Probability Letters*, 73(4):449–457.

Naik, D. N. and Rao, S. S. (2001). Analysis of multivariate repeated measures data with a Kronecker product structured covariance matrix. *Journal of Applied Statistics*, 28(1):91–105.

Roś, B., Bijma, F., de Munck, J. C., and de Gunst, M. C. (2016). Existence and uniqueness of the maximum likelihood estimator for models with a Kronecker product covariance structure. *Journal of Multivariate Analysis*, 143(C):345–361.

Soloveychik, I. and Trushin, D. (2016). Gaussian and robust Kronecker product covariance estimation: Existence and uniqueness. *Journal of Multivariate Analysis*, 149:92–113.

Szczepańska-Álvarez, A., Hao, C., Liang, Y., and von Rosen, D. (2017). Estimation equations for multivariate linear models with Kronecker structured covariance matrices. *Communications in Statistics – Theory and Methods*, 46(16):7902–7915.

Volfovsky, A. and Hoff, P. D. (2015). Testing for nodal dependence in relational data matrices. *Journal of the American Statistical Association*, 110(511):1037–1046.

Wiesel, A. (2012). Geodesic convexity and covariance estimation. *IEEE Transactions on Signal Processing*, 60(12):6182–6189.

Zapata, J., Oh, S. Y., and Petersen, A. (2022). Partial separability and functional graphical models for multivariate Gaussian processes. *Biometrika*, 109(3):665–681.

Zhang, Y., Ekvall, K. O., and Molstad, A. J. (2025). Fast and reliable confidence intervals for a variance component. *Biometrika*, 112(2):asaf010.

Appendix A Proofs

Proof of Proposition 2.3. Let $\Sigma = D(C_2 \otimes C_1)D$ and $\Sigma' = D'(C_2' \otimes C_1')D'$. If $D \neq D'$, then Σ and Σ' have different diagonals, since the diagonal entries of Σ are the squares of those of D.

If $D = D'$ but $C_1 \neq C_1'$, then $\Sigma = \Sigma'$ requires $C_2 \otimes C_1 = C_2' \otimes C_1'$ since D is invertible. The upper-left $r \times r$ block of $C_2 \otimes C_1$ is $[C_2]_{11}C_1 = C_1$, and similarly for $C_2' \otimes C_1'$, so $C_1 \neq C_1'$ gives a contradiction.

If $D = D'$ and $C_1 = C_1'$ but $C_2 \neq C_2'$, then $C_2 \otimes C_1 = C_2' \otimes C_1$ implies $C_2 \otimes I_r = C_2' \otimes I_r$ upon right- and left-multiplying by $I_c \otimes C_1^{-1}$, contradicting $C_2 \neq C_2'$. □

Lemma A.1. *Let $E_i \in \mathbb{R}^{r \times c}$ be the residual matrices defined in Section 2.1, with $n - p \geq \max(r/c, c/r)$, and assume the joint distribution of $Y_1, \ldots, Y_n$ is absolutely continuous. Then, for any positive definite B, almost surely,*

$$\operatorname{rank}\left(\sum_{i=1}^{n} E_i B E_i^{\mathsf{T}}\right) = r;$$

for any positive definite A, almost surely,

$$\operatorname{rank}\left(\sum_{i=1}^{n} E_i^{\mathsf{T}} A E_i\right) = c;$$

and all diagonal elements of $S_n = \sum_{i=1}^{n} \operatorname{vec}(E_i)\operatorname{vec}(E_i)^{\mathsf{T}}$ are almost surely positive.

Proof. Let $\mathcal{Y} = [\operatorname{vec}(Y_1), \ldots, \operatorname{vec}(Y_n)]^{\mathsf{T}} \in \mathbb{R}^{n \times rc}$ and $X = [x_1, \ldots, x_n]^{\mathsf{T}} \in \mathbb{R}^{n \times p}$.

Let $M = I_n - X(X^{\mathsf{T}}X)^{-1}X^{\mathsf{T}}$. Define $E \in \mathbb{R}^{nc \times r}$ by stacking $E_1^{\mathsf{T}}, \ldots, E_n^{\mathsf{T}}$ vertically, so $E = [E_1, \ldots, E_n]^{\mathsf{T}}$; and similarly $\tilde{Y} \in \mathbb{R}^{nc \times r}$ by stacking $Y_1^{\mathsf{T}}, \ldots, Y_n^{\mathsf{T}}$. Then $E = (M \otimes I_c)\tilde{Y}$ and

$$\sum_{i=1}^{n} E_i B E_i^{\mathsf{T}} = E^{\mathsf{T}}(I_n \otimes B)E.$$

Since $I_n \otimes B$ is positive definite, the null spaces of $E^\mathsf{T}(I_n \otimes B)E$ and $E^\mathsf{T}E$ coincide. In particular, $\operatorname{rank}(\sum_{i=1}^n E_i B E_i^\mathsf{T}) = \operatorname{rank}(E)$ for every positive definite B. It suffices to show $\operatorname{rank}(E) = r$.

Write $M = UU^\mathsf{T}$ with $U \in \mathbb{R}^{n\times(n-p)}$ having orthonormal columns. Then $E = (U \otimes I_c)H$, where $H = (U^\mathsf{T} \otimes I_c)\tilde{Y} \in \mathbb{R}^{(n-p)c\times r}$. Since $U \otimes I_c$ has full column rank, $\operatorname{rank}(E) = \operatorname{rank}(H)$.

Now $\operatorname{vec}(H) = (I_r \otimes U^\mathsf{T} \otimes I_c)\operatorname{vec}(\tilde{Y})$. The matrix $I_r \otimes U^\mathsf{T} \otimes I_c$ has full row rank $(n-p)cr$, so this linear map from $\mathbb{R}^{nrc}$ to $\mathbb{R}^{(n-p)cr}$ is onto. Since the entries of $\tilde{Y}$ are those of $Y_1, \ldots, Y_n$, the distribution of $\operatorname{vec}(\tilde{Y})$ is absolutely continuous, and therefore so is that of H. The set of $(n-p)c \times r$ matrices with rank below $\min((n-p)c, r)$ is a proper algebraic subset, hence has Lebesgue measure zero. Therefore $\operatorname{rank}(H) = \min((n-p)c, r)$ almost surely, and since $n-p \geq r/c$ gives $(n-p)c \geq r$, it follows that $\operatorname{rank}(H) = r$ almost surely. The same argument with r and c interchanged shows $\operatorname{rank}\Big(\sum_{i=1}^n E_i^\mathsf{T} A E_i\Big) = c$ almost surely, using $(n-p)r \geq c$ from $n-p \geq c/r$.

For the diagonal entries of S_n, let $\mathcal{Y} \in \mathbb{R}^{n\times rc}$ be the data matrix with i-th row $\operatorname{vec}(Y_i)^\mathsf{T}$ and $\mathcal{E} = M\mathcal{Y}$. Then $[S_n]_{jj} = \sum_{i=1}^n [\operatorname{vec}(E_i)]_j^2$ is the squared norm of the j-th column of $\mathcal{E}$, which is $My_{\cdot j}$, where $y_{\cdot j} \in \mathbb{R}^n$ collects the j-th entry of $\operatorname{vec}(Y_i)$ across $i = 1, \ldots, n$. Since M has rank $n-p \geq 1$, its null space $\operatorname{col}(X)$ is a proper subspace of $\mathbb{R}^n$, so the event $My_{\cdot j} = 0$ has Lebesgue measure zero. The marginal distribution of $y_{\cdot j}$ is absolutely continuous, so $[S_n]_{jj} > 0$ almost surely. □

Proof of Theorem 2.1. The proof proceeds in order. Let $Q_X = I_n - X(X^\mathsf{T}X)^{-1}X^\mathsf{T}$ and $\mathcal{Y} \in \mathbb{R}^{n\times rc}$ have ith row $\operatorname{vec}(Y_i)^\mathsf{T}$, so that $\mathcal{E} = Q_X\mathcal{Y}$ has ith row $\operatorname{vec}(E_i)^\mathsf{T}$.

Part (i). The argument proceeds by induction on k. The base case $k = 0$ holds because $C_1^{(0)} = I_r$ and $C_2^{(0)} = I_c$ are positive definite and $D^{(0)} = I_{rc}$ is in the feasible set. For the inductive step, assume $C_1^{(k)}$ and $C_2^{(k)}$ are positive definite and let $A^{(k)} = (C_2^{(k)} \otimes C_1^{(k)})^{-1}$. With all parameters except d_j fixed, the d_j-dependent terms of ℓ_n are $h(d_j) = -n\log d_j - b_j^{(k)}/(8d_j^2) - a_j^{(k+1)}/d_j$, where the middle term uses $b_j^{(k)} = 4A_{jj}^{(k)}[S_n]_{jj}$ and the last term collects the cross terms $A_{jm}^{(k)}[S_n]_{jm}/d_m$ for $m \neq j$ (using symmetry of $A^{(k)}$ and S_n). The quantity $[S_n]_{jj} > 0$ almost surely by Lemma A.1. Since $A_{jj}^{(k)} > 0$, both $b_j^{(k)} > 0$ and $h(d_j) \to -\infty$ as $d_j \to 0^+$ or $d_j \to \infty$, so h attains its maximum on $(0, \infty)$. Setting $h'(d_j) = 0$ and multiplying by $-d_j^3$ gives

$$nd_j^2 - a_j^{(k+1)}d_j - \tfrac{1}{4}b_j^{(k)} = 0,$$

a quadratic with positive discriminant $(a_j^{(k+1)})^2 + nb_j^{(k)} > 0$ and unique positive root $(a_j^{(k+1)} +$

$\sqrt{(a_j^{(k+1)})^2 + nb_j^{(k)}})/(2n)$, confirming (5).

For $\check{C}_1^{(k+1)}$, the objective over positive definite matrices C_1 is, up to constants,

$$g_1(C_1) = -\frac{nc}{2}\log\det(C_1) - \frac{1}{2}\operatorname{tr}\Big(C_1^{-1}T_1^{(k)}\Big),$$

where $T_1^{(k)} = \sum_{i=1}^n \check{E}_i^{(k+1)}(C_2^{(k)})^{-1}(\check{E}_i^{(k+1)})^\mathsf{T}$. Routine calculations show that g_1 is strictly concave in C_1^{-1}, and hence has unique maximizer $T_1^{(k)}/(nc)$ provided $T_1^{(k)}$ is positive definite. This holds almost surely, as shown next.

Define $\check{E} = [\check{E}_1^{(k+1)}, \ldots, \check{E}_n^{(k+1)}]^\mathsf{T} \in \mathbb{R}^{nc\times r}$ by stacking $(\check{E}_i^{(k+1)})^\mathsf{T}$ vertically. Then $T_1^{(k)} = \check{E}^\mathsf{T}(I_n \otimes (C_2^{(k)})^{-1})\check{E}$. Since $(C_2^{(k)})^{-1}$ is positive definite, the null spaces of $T_1^{(k)}$ and $\check{E}^\mathsf{T}\check{E}$ coincide, so $\operatorname{rank}(T_1^{(k)}) = \operatorname{rank}(\check{E})$. It remains to show $\operatorname{rank}(\check{E}) = r$ almost surely.

Partition $\check{E}$ into c blocks: the bth block $\check{E}^{(b)} \in \mathbb{R}^{n\times r}$ has (i,a) entry $[E_i]_{a,b}/\check{d}_{(b-1)r+a}^{(k+1)}$, so $\check{E}^{(b)} = E^{(b)}D_b^{-1}$, where $E^{(b)} \in \mathbb{R}^{n\times r}$ has (i,a) entry $[E_i]_{a,b}$ and $D_b = \operatorname{diag}(\check{d}_{(b-1)r+1}^{(k+1)}, \ldots, \check{d}_{br}^{(k+1)})$. Since each $\check{d}_j^{(k+1)} > 0$ almost surely, D_b is invertible and $\operatorname{rank}(\check{E}^{(b)}) = \operatorname{rank}(E^{(b)})$. The columns of $E^{(b)}$ are r columns of $\mathcal{E} = Q_X\mathcal{Y}$, so by the argument in Lemma A.1, $\operatorname{rank}(E^{(b)}) = \min(n-p, r) = r$ almost surely; the last equality uses $n-p \geq r$. Since $\check{E}^{(b)}$ is a row-submatrix of $\check{E}$ with r linearly independent columns, $\operatorname{rank}(\check{E}) = r$ almost surely. Note that Lemma A.1 only requires $n-p \geq r/c$ because it uses all nc rows of the stacked matrix, whereas the single-block argument here uses only n rows and therefore requires $n-p \geq r$.

For $\check{C}_2^{(k+1)}$, the objective has the same form with maximizer $T_2^{(k)}/(nr)$, where $T_2^{(k)} = \sum_{i=1}^n (\check{E}_i^{(k+1)})^\mathsf{T}(\check{C}_1^{(k+1)})^{-1}\check{E}_i^{(k+1)}$. Since $\check{C}_1^{(k+1)}$ is positive definite almost surely, the null space argument reduces the rank of $T_2^{(k)}$ to that of the $nr \times c$ matrix F obtained by stacking $\check{E}_1^{(k+1)}, \ldots, \check{E}_n^{(k+1)}$ vertically. Partition F into r blocks: the ath block $F^{(a)} \in \mathbb{R}^{n\times c}$ satisfies $F^{(a)} = G^{(a)}\tilde{D}_a^{-1}$, where $G^{(a)} \in \mathbb{R}^{n\times c}$ has (i,b) entry $[E_i]_{a,b}$ and $\tilde{D}_a = \operatorname{diag}(\check{d}_a^{(k+1)}, \check{d}_{r+a}^{(k+1)}, \ldots, \check{d}_{(c-1)r+a}^{(k+1)})$. By the argument in Lemma A.1, $\operatorname{rank}(G^{(a)}) = \min(n-p, c) = c$ almost surely, using $n-p \geq c$. Since $\tilde{D}_a$ is invertible and $F^{(a)}$ is a row-submatrix of F, $\operatorname{rank}(F) = c$ and $T_2^{(k)}$ is positive definite almost surely.

Part (ii). Consider $C_1^{(k+1)}$ from (9). Since $\check{C}_1^{(k+1)}$ is positive definite (part (i)) and $\check{D}_1^{(k+1)} = (I_r \circ \check{C}_1^{(k+1)})^{1/2}$ is invertible, $C_1^{(k+1)} = (\check{D}_1^{(k+1)})^{-1}\check{C}_1^{(k+1)}(\check{D}_1^{(k+1)})^{-1}$ is positive definite. Its diagonal entries are $[C_1^{(k+1)}]_{jj} = [\check{C}_1^{(k+1)}]_{jj}/[\check{C}_1^{(k+1)}]_{jj} = 1$, so $C_1^{(k+1)}$ is a correlation matrix. The argument for $C_2^{(k+1)}$ is identical. That $D^{(k+1)}$ is a diagonal matrix with positive entries follows directly from (8) and the positivity of $\check{D}^{(k+1)}$, $\check{D}_1^{(k+1)}$, and $\check{D}_2^{(k+1)}$.

Part (iii). Each coordinate update weakly increases ℓ_n by definition of the argmax, and strictly so unless the current iterate already maximizes that coordinate. The rescaling step

(8)–(10) leaves $D(C_2 \otimes C_1)D$ unchanged, hence ℓ_n unchanged. The algorithm terminates when no coordinate update strictly increases ℓ_n, i.e., at a stationary point. $\square$

Appendix B Penalized updates

Let $\bar{D} \in \mathbb{R}^{r\times c}$ be the matrix with (j_1, j_2) entry equal to the standard deviation of response element (j_1, j_2), i.e., $[\bar{D}]_{j_1 j_2} = d_{j_1+(j_2-1)r}$. For fixed $\check{D}^{(k+1)}$ and $C_2^{(k)}$, the penalty satisfies $\mathrm{tr}(\Sigma^{-1}) = \mathrm{tr}(W_r^{(k)} C_1^{-1})$, where $W_r^{(k)} = \mathrm{diag}(\check{\bar{D}}^{-2}\ \mathrm{diag}(C_2^{(k)})^{-1})$ is $r \times r$ diagonal and $\check{\bar{D}}^{-2}$ denotes elementwise squaring and inverting of $\check{\bar{D}}^{(k+1)}$. For fixed $\check{D}^{(k+1)}$ and $\check{C}_1^{(k+1)}$, $\mathrm{tr}(\Sigma^{-1}) = \mathrm{tr}(W_c^{(k)} C_2^{-1})$, where $W_c^{(k)} = \mathrm{diag}((\check{\bar{D}}^{-2})^\mathsf{T}\ \mathrm{diag}(\check{C}_1^{(k+1)})^{-1})$ is $c \times c$ diagonal. The penalized updates replace (6)–(7) with

$$\check{C}_1^{(k+1)} = \frac{1}{nc}\left[\sum_{i=1}^{n} \check{E}_i^{(k+1)} (C_2^{(k)})^{-1} (\check{E}_i^{(k+1)})^\mathsf{T} + \lambda W_r^{(k)}\right], \tag{12}$$

$$\check{C}_2^{(k+1)} = \frac{1}{nr}\left[\sum_{i=1}^{n} (\check{E}_i^{(k+1)})^\mathsf{T} (\check{C}_1^{(k+1)})^{-1} \check{E}_i^{(k+1)} + \lambda W_c^{(k)}\right], \tag{13}$$

and the update for $\check{d}_j$ in (5) is unchanged in form, with $[S_n]_{jj}$ replaced by $[S_n]_{jj} + \lambda/n$. The rescaling step (8)–(10) is unchanged.

Appendix C Fisher information

Without loss of generality, assume that $\mathcal{B} = 0$ since the score for the covariance parameters depends on $\mathcal{B}$ only through $\mathrm{vec}(Y_i) - \mathcal{B}^\top x_i \sim \mathcal{N}(0, \Sigma)$, $i \in \{1, \ldots, n\}$. In more general notation, let Σ be parameterized by $\theta = (\theta_1, \ldots, \theta_m)$, where θ_j is either an off-diagonal entry of C_1 or C_2, or a diagonal entry of D. The log-likelihood is, up to a constant,

$$\ell_n(\theta) = -\frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{n} \mathrm{vec}(Y_i)^\mathsf{T} \Sigma^{-1}\, \mathrm{vec}(Y_i).$$

With $H_j = \partial\Sigma/\partial\theta_j$, the score is

$$\frac{\partial \ell_n}{\partial \theta_j} = -\frac{n}{2}\,\mathrm{tr}(\Sigma^{-1} H_j) + \frac{1}{2}\sum_{i=1}^{n} \mathrm{vec}(Y_i)^\mathsf{T} \Sigma^{-1} H_j \Sigma^{-1}\, \mathrm{vec}(Y_i).$$

The Fisher information has elements $\mathcal{I}_{jk} = \mathrm{cov}(\partial \ell_n/\partial\theta_j,\, \partial \ell_n/\partial\theta_k)$. The first term in the score is constant in the data and does not contribute; independence of observations gives

$$\mathcal{I}_{jk} = \frac{n}{4}\,\mathrm{cov}\Big(\mathrm{vec}(Y_1)^\top A_j\,\mathrm{vec}(Y_1),\ \mathrm{vec}(Y_1)^\top A_k\,\mathrm{vec}(Y_1)\Big)\,,$$

where $A_j = \Sigma^{-1}H_j\Sigma^{-1}$ and $\mathrm{vec}(Y_1) \sim \mathcal{N}(0,\Sigma)$. For a random vector $v \sim \mathcal{N}(0,\Sigma)$ and symmetric matrices A, B, the identity $\mathrm{cov}(v^\top Av,\, v^\top Bv) = 2\,\mathrm{tr}(A\Sigma B\Sigma)$ yields

$$\mathrm{cov}\Big(\mathrm{vec}(Y_1)^\top A_j\,\mathrm{vec}(Y_1),\ \mathrm{vec}(Y_1)^\top A_k\,\mathrm{vec}(Y_1)\Big) = 2\,\mathrm{tr}(\Sigma^{-1}H_j\Sigma^{-1}H_k),$$

and therefore $\mathcal{I}_{jk} = \frac{n}{2}\,\mathrm{tr}(\Sigma^{-1}H_j\Sigma^{-1}H_k)$.

Computation of the information matrix is efficient because $\Sigma^{-1} = D^{-1}(C_2^{-1}\otimes C_1^{-1})D^{-1}$ and each H_j is sparse or has Kronecker structure. For $\theta_j = [C_1]_{ab}$, the derivative is $H_j = D\{C_2\otimes(E_{ab}+E_{ba})\}D$, where E_{ab} is the elementary matrix with a one in position (a,b) and zeros elsewhere. The mixed-product property gives $\Sigma^{-1}H_j = D^{-1}\{I_c\otimes C_1^{-1}(E_{ab}+E_{ba})\}D$, so for $\theta_k = [C_1]_{cd}$,

$$\mathrm{tr}(\Sigma^{-1}H_j\Sigma^{-1}H_k) = c\cdot\mathrm{tr}\Big\{C_1^{-1}(E_{ab}+E_{ba})C_1^{-1}(E_{cd}+E_{dc})\Big\}\,.$$

For a cross pair $\theta_j = [C_1]_{ab}$ and $\theta_k = [C_2]_{cd}$, the trace factors as $\mathrm{tr}\{C_1^{-1}(E_{ab}+E_{ba})\}\cdot\mathrm{tr}\{C_2^{-1}(E_{cd}+E_{dc})\}$. For $\theta_j = d_k$, $H_j = e_ke_k^\top(C_2\otimes C_1)D + D(C_2\otimes C_1)e_ke_k^\top$ has rank two, so the trace reduces to a sum of $O(rc)$ entries of Σ^{-1}. In each case, the $rc\times rc$ matrices need not be formed explicitly.